ASHESI UNIVERSITY

INTRO TO ARTIFICIAL INTELLIGENCE FINAL PROJECT

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Group 1

# Automated Essay Scoring and Feedback Generating System

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**Abstract**

Grading essays by hand takes a lot of time, and different teachers may give slightly different scores to the same essay. Students also need more than just a score; they need to know what they did well and how they can improve.

This project developed an Automated Essay Scoring (AES) system that gives English essays a score and provides useful feedback. The system was trained using the ASAP 2.0 dataset, which contains 24,728 essays from seven different writing prompts.

Several features were taken from each essay, including word and sentence counts, vocabulary variety, grammar information, parts of speech, and readability. Three machine learning models were tested: Ridge Regression, Gradient Boosting, and Random Forest. Gradient Boosting gave the best results, with a Quadratic Weighted Kappa (QWK) score of 0.74 and a Mean Absolute Error (MAE) of 0.48 on the test data. This shows that the system's scores were fairly close to the scores given by human graders.

The system also includes a rule-based feedback component. It looks at the essay's features and gives simple comments about its strengths, weaknesses, and areas for improvement. In this way, the system not only provides a score but also explains the result and gives students useful guidance.

## 1. Introduction

**1.1. The Problem :** Grading essays can be one of the most time-consuming tasks for teachers. Grading a large number of essays can take several hours, especially when teachers need to make sure that every student is graded fairly and consistently. Another problem is that teachers may give different scores to the same essay. Teachers can have different opinions about the quality of a student's writing. Students also need useful feedback, not just a final score. They need to understand what they did well, what they did poorly, and what they should do differently next time. However, because teachers have limited time, feedback is sometimes short, general, or not provided at all.

**1.2. Why It Matters :** Automated Essay Scoring can help solve some of these problems. Instead of a teacher manually grading every essay, a computer can analyze the writing and provide a score quickly and consistently. However, giving students only a score is not enough. A student who receives a score of 6, for example, may still not know why they received that score or how to improve.

For this reason, this project combines automatic scoring with explainable feedback. The goal is to help teachers save time while still giving students useful information about their writing.

**1.3. Objectives :** The main goals of this project were to:

- Predict essay scores using essays that have already been graded by human teachers.
- Analyze important features of writing, such as vocabulary, grammar, sentence structure, readability, and essay length.
- Generate simple and understandable feedback that explains the strengths and weaknesses of an essay.
- Measure how well the system performs using appropriate evaluation measures, especially Quadratic Weighted Kappa (QWK), which is commonly used for essay scoring.

## 2. Background

**2.1. Automated Essay Scoring :** Automated Essay Scoring has been studied for many years. One of the early systems was Project Essay Grade (PEG), which used basic features such as essay length and word complexity to estimate essay scores. Over time, AES systems have become more advanced. Most approaches can be grouped into three main types.

**Feature-based models** use information from the essay, such as word count, sentence length, vocabulary variety, readability, and grammar. These features are then given to machine learning models such as regression, Random Forest, or Gradient Boosting. These systems are relatively fast and easier to understand, but their performance depends on the quality of the features used.

**Deep learning models** use methods such as BERT and other transformer models. Instead of manually selecting features, these models learn patterns directly from the essay text. They can achieve strong results, but they usually require more computational power and are difficult to explain.

**Hybrid systems** The Automated Student Assessment Prize (ASAP) dataset is one of the most commonly used datasets for research in automated essay scoring. It provides a useful benchmark for testing different AES approaches. This project uses the ASAP 2.0 dataset.

**2.2. Feedback Generation :** Compared with automatic scoring, automatic feedback has received less attention. Some systems use simple rules to generate feedback. For example, if an essay contains very long sentences, the system might suggest using shorter sentences. If the essay has a limited vocabulary, it might recommend using a wider range of words.

More recent systems use Large Language Models (LLMs) to produce more natural and personalized feedback. Although these systems can produce fluent responses, it can be difficult to understand exactly why a particular piece of feedback was given.

This project uses a rule-based feedback system. This approach was chosen because it is easier to understand and explain. Each piece of feedback is connected to a specific feature of the essay. For example, a comment about grammar can be linked directly to the number of grammar errors found in the essay.

**2.3. Positioning of This Project :** This project uses a traditional machine learning approach because it provides a good balance between performance, speed, and explainability. The system looks at more than basic features such as essay length. It also considers readability, vocabulary diversity, parts of speech, grammar errors, and other characteristics of writing. The scoring model is combined with a rule-based feedback system. This means the final system provides both a predicted score and an explanation of how the student can improve.

The main idea is simple: the system should not just tell a student "Here is your score." It should also help answer "Why did I get this score, and what can I do better?"combine traditional features with deep learning methods. The goal is to get the benefits of both approaches, such as good performance and better explainability.

**3. Methodology:** The system was developed using the ASAP 2.0 (Automated Student Assessment Prize) dataset, obtained via Kaggle (lburleigh/asap-20). The working dataset consisted of 24,728 student essays, each associated with an essay_id, a human-assigned score, the essay text ( full_text), and metadata identifying the writing prompt (prompt_name) and assignment. Essays span multiple distinct prompts, so prompt identity was retained and later one-hot encoded rather than discarded, since score expectations can differ across prompts.

**3.1. Models Used :** Three different regression models were trained and compared:

- Ridge Regression : a linear model that includes regularization to reduce overfitting. The regularization value was set to 1.0. The features were standardized before training because Ridge Regression can be affected by differences in feature scales.
- Gradient Boosting Regressor : a model that uses several decision trees one after another. Each new tree tries to correct the mistakes made by the previous trees. The model used 100 trees, a maximum tree depth of 3, and a learning rate of 0.1.
- Random Forest Regressor : a model that builds many decision trees independently and combines their predictions. The model used 100 trees.

**3.2. Exploratory data analysis :** Before we performed our feature engineering or modelling decisions we did some exploratory data analysis to learn more about the dataset and this informed our design decision.

We first printed out the statistical information about the dataset and the general information about it, the columns they contain, the datatype for each particular column, the maximum and minimum score in the table and many other statistical information.

We visualised some of our features to get the variations of the dataset, we first visualised the score distribution and we realised that the scores were not evenly spread across the 1 - 6 scale most scores were centered around 2 - 4 range with about 9021 essays scoring 3, 6847 essays scoring 2, and 5553 essays scoring 4. This imbalance was treated as a design signal rather than a footnote, a model trained on this distribution will be more likely to regress its predictions toward the middle of the scale. Essay length distribution was also visualised to get the distribution of the essays length and it was noticed that many essays had long length.  We also visualised the prompt distribution and the observation that was made motivated a decision to noe-hot-encode prompt_name.

**3.3. Feature selection:** Out of all the columns in the dataset we decided to keep only five columns for our training because most of the drop columns contained a lot of null values e.g source_text_1 and until source_text_4 and this columns were basically the same information from the assignment, some columns were also dropped because it will not add any meaningful analysis to the evaluation but rather introduce some bias into the model and these columns are gender, disability, economically_disadvantage and race_ethnicity. After dropping these columns we were left with essay_id, score, full_text, assignment, and prompt_name.

Aside the features that was gotten from the dataset we also added our features to give the model the maximum information it needs to make the right predictions, and the features we added were; readability formulas (Flesch Reading Ease, Flesch-Kincaid, Gunning Fog, Dale-Chall), type-token ratio (TTR), part-of-speech ratios (conjunction ratio, preposition ratio), word_count, average word count and many more summing up to about 18 basic features.

**3.4. Feature Engineering:** We used pandas vectorized string operations and .apply() with custom python functions to add the features we encoded onto the dataframe we will be working with for the rest of the project. “texstat” was used for all the four readability formulas rather than re implementing them, scikit-learns ENGLISH_STOP_WORDS was used for stopword-ratio calculation, keeping stopword handling consistent with the same library ecosystem already used for modelling and scaling, and spaCy (en_core_web_sm) was used for part-of-speech tagging and named-entity recognition.

The 18 basic, vocabulary, readability, and grammar-ratio features were computed for every essay and fed directly into the three scoring models. POS/named-entity extraction, YAKE keyword extraction, and LanguageTool grammar-error counting was also computed across the dataset, but reserved for the feedback-generation stage.

Part-of-speech tagging used spaCy's pre-trained pipeline, but not its full default configuration, the parser and lemmatizer components were explicitly disabled at load time, since neither contributes to part-of-speech counts or named-entity recognition. Lexical diversity was computed using a moving-average type-token ratio (MATTR) rather than a single whole-essay ratio. Keyword extraction used YAKE, an unsupervised, statistics-based method rather than a trained model. Grammar-error counting used LanguageTool, a rule-based grammar checker that runs as a local server process and evaluates text against a large set of pattern-matching grammar and style rules, returning a count of flagged issues per essay. Named-entity counts were included as a proxy for how concretely an essay supports its argument .

**4. Implementation**

**4.1. System Architecture:** The system was built as a four-stage process: data preprocessing, feature extraction, model training, and evaluation. After the model produces a score, the result is passed to a feedback module.

**4.2. Data Splitting** : To make sure the evaluation was fair, the dataset was divided into three groups instead of using only training and testing data:

- Training set: 14,836 essays (60%)
- Validation set: 4,946 essays (20%)
- Test set: 4,946 essays (20%)

The training set was used to train the models. The validation set was used to compare the models and make development decisions. The test set was kept separate and was only used for the final evaluation. This approach helps avoid data leakage, where information from the test data could accidentally influence the model during development.

**4.3. Features Used** : For each essay, 18 features were created and used as input for the models. These features describe different aspects of the essay, including:

- Basic writing statistics: word count, character count, number of sentences, number of paragraphs, average word length, average sentence length, and average paragraph length
- Readability: Flesch Reading Ease, Flesch-Kincaid Grade Level, Gunning Fog Index, and Dale-Chall Readability Score
- Vocabulary: type-token ratio, Guiraud index, stopword ratio, and long-word ratio
- Grammar and sentence structure: conjunction ratio and other ratios based on parts of speech

Some essays produced missing or infinite values, for example, when an essay had no sentences. These values were replaced using the median values from the training set. The same values were then used for the validation and test sets. This helped prevent information from the test data from affecting the model.

A grid search was also carried out for the Random Forest model. Different numbers of trees, tree depths, and minimum numbers of samples per leaf were tested to see whether tuning the model could improve its performance. Because essay scores are whole numbers, the models' continuous predictions were rounded to the nearest whole number when calculating accuracy and Quadratic Weighted Kappa (QWK).

## 5. Results and Evaluation

**5.1. Evaluation Metrics :** Four metrics were used to evaluate the models on the test set:

- MSE (Mean Squared Error): Measures the average squared difference between the predicted and actual scores. Bigger mistakes are penalized more heavily.
- MAE (Mean Absolute Error): Shows, on average, how many score points the prediction differs from the actual score.
- $R^2$ Score: Shows how much of the variation in essay scores the model can explain.
- QWK (Quadratic Weighted Kappa): Measures how closely the model's scores agree with human-assigned scores. It also gives more weight to larger scoring differences. QWK is commonly used in automated essay scoring and was the main metric used in this study.

**Model Comparison**

| Model | MSE | MAE | R^2 | Accuracy | QWK |
|---|---|---|---|---|---|
| Gradient Boosting | 0.401 | 0.480 | 0.624 | 0.624 | 0.739 |
| Random Forest | 0.412 | 0.487 | 0.613 | 0.613 | 0.733 |
| Ridge Regression | 0.431 | 0.500 | 0.595 | 0.595 | 0.716 |

Gradient Boosting gave the best results across all four evaluation measures. It achieved a QWK score of 0.74 and an MAE of 0.48. This means that, on average, its predictions were less than half a point away from the human-assigned score. The QWK result also shows that the model has a reasonably strong level of agreement with human scores. However, it is important to note that the model is not a replacement for human marking.

The three models performed fairly similarly, with QWK scores ranging from 0.72 to 0.74. This suggests that the features used in the system may have had a bigger impact on performance than the choice of model. In other words, improving the features could potentially lead to larger improvements than simply changing or tuning these three models. A more advanced approach, such as using a transformer model trained directly on the essay text, could also be explored in future work.

### 5.2. Sample Predictions

The table below shows ten example predictions from the test set:

| Actual Score | Predicted Score |
|---|---|
| 3 | 3.79 |
| 2 | 2.12 |

| | |
|---|---|
| 2 | 2.24 |
| 3 | 3.89 |
| 2 | 1.88 |
| 4 | 3.54 |
| 1 | 3.30 |
| 2 | 2.02 |
| 3 | 3.52 |
| 5 | 3.52 |

Most of the predictions are fairly close to the actual scores. However, there are some noticeable differences. The largest difference appears in the seventh example. The actual score is 1, while the model predicted 3.30. This type of error could happen when an essay is very short, does not properly answer the prompt, or has writing characteristics that are different from most of the essays used to train the model.

This type of example is useful because it shows where the model could be improved.

**5.3. Feature Importance :** The feature importance results show which features had the biggest influence on the Gradient Boosting model's predictions. Character count was by far the most important feature, accounting for about 58% of the total feature importance. Other important features included average paragraph length, Dale-Chall readability, conjunction ratio, and long-word ratio. These other features each contributed around 3–4%.

The strong influence of character count is an important limitation of the system. It suggests that essay length may have a large effect on the predicted score. This is a common concern in automated essay scoring. A longer essay may receive a higher predicted score even when its actual writing quality is not necessarily better. Therefore, the model may be partly learning that "longer essays tend to score higher" rather than fully understanding the quality and meaning of the writing. This issue should be considered when interpreting the model's results and is discussed further in the limitations and ethical considerations sections.

**5.4. Hyperparameter Tuning :** A grid search was performed on the Random Forest model to see whether different settings could improve its performance.
The search tested:

- Number of trees: 100 and 200
- Maximum tree depth: None, 10, and 20
- Minimum samples per leaf: 1, 2, and 5

The purpose of this experiment was to find out whether a better Random Forest configuration could outperform the original model. If the tuned model performed only slightly better than the original model, this would suggest that the baseline Random Forest was already performing close to its best with the selected features. If the improvement was large, the tuned configuration should be reported as the final Random Forest setup.

**6. Ethical Considerations** : Bias and fairness are critical issues in automated essay scoring. Since the dataset includes demographic attributes such as race, gender, and disability status, there is a risk that models may unintentionally learn patterns that disadvantage certain groups. This could lead to unfair scoring outcomes if not carefully monitored. Because of this unfairness that might occur in the model we remove such features from the dataset we trained and tested.

Privacy is equally important, as student essays often contain personal information. Ensuring anonymization, secure storage, and compliance with data protection standards is necessary to protect students.

Transparency is another challenge: while models like Gradient Boosting and Random Forest achieve strong predictive performance, their complexity makes it difficult for teachers and students to fully understand how scores are derived.

To build trust, the system should incorporate explainability tools and clear documentation that link predictions to specific essay features.

## 7. Limitations and Future Work

**7.1. Weakness:** The main weakness is that the final score depends heavily on the AI model's prediction. Although the model performs well, with a Quadratic Weighted Kappa (QWK) of about 0.74, it can still make mistakes or show bias because it was trained on specific types of essays. A strong essay could receive a low score, while a weaker essay might receive a high score if it matches patterns the model associates with good writing. This highlights the need to treat AI scores as recommendations for teacher review rather than final grades. Human oversight ensures fairness and prevents over-reliance on automated scoring.

**7.2. Future Improvement** : A possible improvement is to add an AI-generated summary of each essay. Instead of requiring teachers to read the entire essay again, the system could provide a short summary of the student's main ideas and arguments. This would allow teachers to quickly check whether the predicted score and feedback make sense, while still maintaining human oversight. By combining predicted scores, feedback, and summaries, the system would act as a teacher-support tool rather than a replacement for human judgment.

**7.3. Future Work** : Future work could explore multilingual essay scoring to make the system more inclusive for diverse classrooms. Testing the system in real educational settings would provide valuable insights into usability and fairness. Additionally, integrating human-in-the-loop review and feedback mechanisms would strengthen trust in the system and ensure that students receive constructive guidance alongside their scores. Finally, combining traditional feature-based approaches with deep learning methods could improve performance while maintaining explainability.

**Conclusion:** This project demonstrated that traditional machine learning models such as Ridge Regression, Gradient Boosting, and Random Forest can predict essay scores with reasonable accuracy, achieving QWK values around 0.74 and MAE of 0.48. The inclusion of a rule-based feedback component ensures that students receive not only a score but also practical guidance on how to improve their writing. While the system shows promise in reducing teacher workload and providing consistent scoring, ethical risks around bias, fairness, and privacy mean it should complement not replace human judgment. With broader datasets, transparency tools, and fairness safeguards, the system can evolve into a reliable educational support tool that balances efficiency with accountability.

APPENDICE

**AI Use Declaration Form**

**Course:** Introduction to Artificial Intelligence

**Lab Title:** Final Project

**Students Group:** Group 1_________________________________________

**Student ID:** ____________________________________________________

**GitHub Repository Link:** AdamaBaba/Project

**Date Submitted:** 17/08/2026_________________________________________

**1. AI Use Summary**

| **Question** | **Student Response** |
|---|---|
| Did you use any AI tool for this project? | **Yes** |
| Estimated percentage of the work influenced by AI | 25% |
| Did you attach evidence of AI use where applicable? | **Yes** |

**2. Details of AI Use**

Complete the table below for each AI tool used. Add more rows where necessary.

| **Name of AI Tool Used** | **Purpose for Using the Tool** | **Prompt or Instruction Given to the Tool** | **Part(s) of the Work Influenced by the Tool** | **How we Verified, Edited, or Improved the AI Output** |
|---|---|---|---|---|
| Claude AI | To help generate some codes | | Small parts of each section | |
| Claude AI | To help spot and correct the errors I was getting when running my program | Paste the errors in the ai to just go though it | When running each of our code cells | Make sure it doesn't change our original code, but just tell us what the problem is and how we should correct it. |

| | | | | |
|---|---|---|---|---|
| Claude AI | To help with the designing of the application | | | |

### 3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

| **Evidence Type** | **Attached? (Yes, No, N/A)** | **File Name** |
|---|---|---|
| AI-generated draft | Yes | |
| Prompt history | Yes | |
| Screenshot of AI interaction | Yes | |
| Exported AI conversation | | |
| AI-generated code snippet | Yes | |
| Revised version showing student input | No | |

### 4. Student Declaration

We declare that:

| **Declaration Statement** | **Response (Yes/No)** |
|---|---|
| The submitted work is our own work. | Yes |
| Any use of AI tools has been clearly declared in this form. | Yes |
| The prompts, instructions, and AI-generated outputs have been disclosed where applicable. | No |
| We reviewed, tested, edited, and improved any AI-generated content before submission. | Yes |
| Our AI usage does not exceed 25% of the entire work. | Yes |
| We understand that undeclared or excessive AI use may be treated as academic misconduct. | Yes |

**Student Signature:** ADAMABABA, MARFOMAVIS, MARIAMGOGE, BONIPHACEBENJAMIN

**Date:** 17/08/2026____________________________________________

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